

Mahmoud Hussien

📍 Cairo, Egypt

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Profile

Applied **Machine Learning Engineer** with strong software engineering and cloud background (8+ years).

Experienced in building **end-to-end ML pipelines** for tabular problems including data preprocessing, model selection, cross-validation, and business-aware evaluation.

Experienced in delivering **production-ready ML systems**.

Core Skills

Machine Learning

- Classification & regression (tabular data)
- Feature engineering
- Model comparison & cross-validation
- Business-aware metrics & threshold tuning
- Scikit-learn pipelines
- LightGBM, Logistic Regression
- Familiarity with pre-trained deep learning models

Python & Data

- Python (production pipelines)
- pandas, numpy
- SQL (advanced queries)
- ML project structuring

Cloud & MLOps

- AWS (EC2, S3, Lambda)
- Model artifact saving
- Config-driven ML systems
- Logging & reproducibility
- Basic MLOps practices (experiment tracking, model versioning, inference pipelines)

Software Engineering

- Git (branches, PRs, reviews)
 - Modular architecture
 - CI/CD concepts
 - Agile / Scrum
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Machine Learning Projects

Customer Churn Prediction — End-to-End ML System

GitHub: <https://github.com/mahmoud-hussien12/ML-Training>

- Built a **production-oriented ML pipeline** to predict customer churn probability.
- Implemented clean data loading, stratified train/validation/test splits.
- Designed preprocessing using ColumnTransformer and sklearn pipelines.
- Compared **Logistic Regression vs LightGBM** using stratified cross-validation (ROC-AUC).
- Selected final model based on **validation performance and business trade-offs**.
- Tuned decision thresholds to balance churn recall vs false positives.
- Extracted and interpreted **top features** for business insight.
- Saved final model artifact for inference and deployment.
- Demonstrated how churn prediction can support CRM actions such as targeted retention offers and proactive customer support.

Tech: Python, scikit-learn, LightGBM, pandas, joblib

NLP Product Similarity System (Omdena)

- Worked on an NLP project for product similarity in an online marketplace.
 - Implemented data collection via web scraping (~275k products).
 - Participated in preprocessing and similarity modeling.
 - **Led deployment phase** using AWS EC2 and Streamlit.
 - Focused on improving product similarity relevance over sentiment accuracy.
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Professional Experience

Software Engineer

CamelCase Company — Cairo, Egypt

Feb 2018 – Present

- 8+ years building and maintaining production systems with strong focus on reliability, scalability, and clean architecture.
 - Designed modular backend systems and APIs.
 - Wrote advanced SQL queries used for analytics and insights.
 - Worked closely with clients to translate business needs into technical solutions.
 - Used AWS services (Lambda, S3, CloudFront) in production systems.
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Machine Learning Engineer (Project-Based)

Omdena — Bangladesh Local Chapter

Oct 2021 – Nov 2021

- Collaborated in an end-to-end ML pipeline for an AI-driven marketplace.
- Built data pipelines, preprocessing steps, and similarity models.
- Took ownership of deployment and cloud setup.

Education & Certifications

- **B.Sc. in Computer Science** — Cairo University
 - **AWS Certified Machine Learning – Specialty**
 - **Advanced Data Analysis Nanodegree** — Udacity
 - **Deep Learning Specialization** — Coursera (3 / 5)
 - **Structuring Machine Learning Projects** — Coursera
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Languages

- Arabic: Native
 - English: Professional working proficiency
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